Symphonym: Universal Phonetic Embeddings for Cross-Script Toponym Matching via Teacher-Student Distillation

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Abstract

Linking place names across historical sources, languages, and writing systems remains a fundamental challenge in digital humanities and geographic information retrieval. Existing approaches either require language-specific phonetic algorithms, expert-curated transliteration rules, or fail to capture the phonetic relationships that make "Moscow", "Москва", and "موسکو" recognisably the same place. The objective of this research is to present **Symphonym**, a neural embedding system that maps toponyms from any script into a unified 128-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion or language-specific resources. Symphonym uses a Teacher-Student architecture: a Teacher network trained on articulatory phonetic features (derived via hybrid grapheme-to-phoneme conversion combining Epitran with 100 new language-script extensions, Phonikud for Hebrew, and CharsiuG2P for CJK languages, followed by PanPhon feature extraction) produces target embeddings, while a Student network learns to approximate these embeddings directly from characters. The Student operates at inference time without requiring phonetic resources, enabling deployment at scale. Training completed February 2026 on NVIDIA L40S GPUs using 20.4 million samples from co-located toponym pairs extracted from GeoNames, Wikidata, and the Getty Thesaurus of Geographic Names, with phonetic similarity filtering to exclude false cognates. The system is optimised for *cross-script* matching—linking "北京" to "Beijing"—where traditional string metrics fail entirely. In v6, CJK characters, Hebrew, Greek, Armenian, and other scripts are processed in their native writing systems (not romanized), with IPA transcriptions generated through specialized G2P methods appropriate to each script: CharsiuG2P for Chinese topolects and Korean, Phonikud for Hebrew, and extended Epitran support for Greek, Armenian, and 100+ additional language-script pairs. The system achieves 93% accuracy on cross-script equivalents and 87.5% Recall@1 on the MEHDIE Hebrew-Arabic benchmark, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines. Production deployment tests on 66.9 million toponyms demonstrate 100% embedding coverage across all 20 scripts, validating the hybrid G2P pipeline integration. Symphonym effectively addresses the challenge of cross-script toponym matching by providing a phonetically grounded embedding space, demonstrating strong performance on key benchmarks. Its deployment in the World Historical Gazetteer (WHG) enables phonetic search and reconciliation across 67M+ toponyms, offering a scalable solution for digital humanities research. Beyond toponyms, a case study on medieval personal name matching in the London Customs Accounts (University of London) demonstrates that Symphonym embeddings transfer effectively to historical orthographic variation in single-script contexts, contributing as the 4th-ranked feature (of 11) in a supervised name-clustering pipeline with F1=0.752. This confirms Symphonym's broader applicability to onomastic research and historical document analysis.

1 Introduction

Historical gazetteers aggregate place references from sources spanning millennia, dozens of languages, and multiple writing systems. The World Historical Gazetteer[1] (WHG) alone indexes over 112 million toponym records from antiquity to the present, drawn from sources as diverse as classical Latin itineraries, medieval Arabic geographies, and modern national mapping agencies. Each toponym record is tagged with language and script metadata, though the same surface form (e.g., "London") may appear multiple times when attested in different languages. Linking these references—determining that "Londinium", "Londres", "Лондон", and "伦敦" all denote the same city—is essential for cross-cultural historical research but remains technically challenging.

The core difficulty is that place name similarity is fundamentally *phonetic*, not orthographic. English speakers recognise "Munich" and "München" as the same city not because the spellings match, but because the sounds are similar enough to suggest common origin. Yet computational approaches to toponym matching have largely relied on string-based metrics (edit distance, Jaro-Winkler) or phonetic algorithms designed for single languages (Soundex, Metaphone for English; Cologne phonetic for German). These approaches fail catastrophically when names cross script boundaries: no amount of edit distance tuning will link "東京" to "Tokyo".

Recent work has applied neural embeddings to geographic entity matching, but existing systems either operate within single scripts, require language identification at inference time, or depend on curated transliteration resources that don't

[1] https://whgazetteer.org

scale to the long tail of historical languages. Until now, there has been no system that can place “Москва”, “Moscow”, “موسکو”, and “モスクワ” (Japanese Katakana) near each other in embedding space using only the raw character sequences as input.

This paper presents such a system, which I call *Symphonym*. The key contributions are:

1. A **script-aware but script-agnostic embedding architecture** that maps toponyms from 20+ writing systems into a unified 128-dimensional space where proximity reflects phonetic similarity.

2. A **Teacher-Student training framework** that transfers phonetic knowledge from articulatory features (available only for IPA-transcribed names) to a character-level model requiring no phonetic resources at inference time.

3. A **noise-aware training regime** with strategic language dropout that forces the model to learn script-intrinsic phonetic patterns rather than relying on language-specific shortcuts.

4. **Phonetic similarity filtering** for training data that prevents the model from learning false equivalences between unrelated exonyms (e.g., “Germany” $\neq$ “Deutschland”).

5. A **three-phase curriculum** progressing from phonetic feature learning through knowledge distillation to hard negative discrimination.

6. **Empirical validation** of cross-domain transfer: a case study on medieval personal name matching (Section 8) demonstrates that embeddings trained on modern toponyms transfer effectively to historical personal names with pre-standardisation orthography, contributing substantially to a supervised name-clustering pipeline.

The resulting system will enable fuzzy phonetic search across the full WHG corpus without language identification, script-specific preprocessing, or runtime phonetic conversion. Beyond cross-script matching, the phonetic embedding approach naturally handles *historical orthographic variation*—the inconsistent spellings characteristic of pre-standardisation documents—by grouping sound-alike variants near their modern canonical forms. This capability extends beyond toponyms to personal names, archival records, and any textual domain involving phonetically consistent but orthographically variable name forms.

Scope and Integration. This work addresses the *name-matching* component of toponym resolution specifically. The model has no access to geographic coordinates or spatial context—it operates purely on phonetic similarity between strings. Both Symphonym and WHG’s new reconciliation pipeline (termed WHG

PLACE: Place Linkage, Alignment, and Concordance Engine) build on architecture developed by Grossner and Mostern (2021). Within the Place Linkage, Alignment, and Concordance Engine (PLACE), phonetic similarity serves as one input to a larger process: candidate toponyms are first retrieved via the phonetic-aware toponym index, then places containing matching toponyms are selected from the places index and filtered by geographic proximity and other contextual criteria. The contribution here is the phonetic similarity component alone.

In practical terms, Symphonym enhances WHG's Reconciliation Service API[2] (v2.0), facilitating the work of digital humanities and GLAM (galleries, libraries, archives, museums) professionals who seek to link place references in historical documents, archival catalogues, and research datasets to WHG's consolidated index. Once linked, these references become entry points to a web of related materials—other sources attesting the same places, variant historical names, and cross-references across collections—even when the original materials span multiple scripts, languages, and historical periods.

2 Related Work

2.1 Classical Phonetic Algorithms

The earliest computational approaches to phonetic matching emerged from English-language record linkage. Soundex (Russell, 1918), developed for US Census indexing, maps names to four-character codes by retaining the first letter and encoding subsequent consonants. Its English-centricity is explicit: the algorithm assumes Latin script and English phoneme-grapheme correspondences. Gadd (Gadd, 1990) addressed some of Soundex's limitations with PHONIX, developed for the URICA library system in multilingual Southern Africa. Gadd hand-crafted 160 letter-group transformation rules to standardise spellings before phonetic coding—for example, mapping the Afrikaans-influenced cluster “tj” followed by a vowel to “ch”, so that variant transliterations of names like Chekhov (TSJECHOF, TJEKHOW, CHEKHOV) would converge to a common code. PHONIX represents the high-water mark of the rule-based approach: a single developer's linguistic intuition encoded as explicit transformations. The contrast with the present work is instructive: where Gadd wrote 160 rules by hand for Latin-script names, Symphonym's Teacher network learns phonetic relationships from 20 million training samples across 20 writing systems, discovering correspondences that would be impractical to enumerate manually. Metaphone (Philips, 1990) and Double Metaphone (Philips, 2000) improve accuracy for English by handling common spelling variations, but remain fundamentally tied to English phonology.

Language-specific variants exist for other European languages, but each requires separate implementation and fails outside its target language family. No

[2] https://docs.whgazetteer.org/content/technical/apis.html#reconciliation-service-api

classical algorithm handles cross-script matching: they assume a single alphabet and a known phoneme inventory.

2.2 String Similarity Metrics

Edit distance metrics (Levenshtein, Damerau-Levenshtein, Jaro-Winkler) measure orthographic rather than phonetic similarity. While useful for detecting OCR errors or spelling variants within a script, they cannot capture phonetic relationships across scripts. The Levenshtein distance between "Moscow" and "Москва" is undefined without transliteration; even after romanisation to "Moskva", the distance of 2 fails to reflect the near-identical pronunciation.

Recchia and Louwerse (Recchia and Louwerse, 2013) evaluated 21 similarity measures on romanised toponyms from 11 countries, finding that optimal measures varied by language but were consistent within linguistic families. Santos et al. (Santos et al., 2018) demonstrated that combining multiple string metrics with machine learning classifiers improved matching accuracy for Portuguese toponyms.

Phonetic-aware edit distances (Kondrak, 2000) weight substitutions by articulatory similarity, but still require input in a common alphabet.

2.3 Neural Approaches to Entity Matching

Recent work applies neural embeddings to geographic entity resolution. Qiu et al. (Qiu et al., 2024) proposed GeoBERT, combining a BERT model pre-trained on Chinese geographic text with the Enhanced Sequential Inference Model (ESIM) architecture for toponym pair classification. Their approach integrates multiple Chinese-specific features including Pinyin romanisation, Wubi keyboard encoding, character radicals, and stroke counts. On Chinese address matching datasets, GeoBERT achieved F1 scores exceeding 0.97. However, the Chinese-specific features have no analogues in other writing systems, and the geographic pre-training corpus is monolingual, limiting cross-lingual application.

Most relevant is the MEHDIE project (Sagi et al., 2025), which developed tools for matching toponyms between historical Hebrew and Arabic sources. Sagi et al. present a benchmark comprising place pairs from 13th-century Arabic geographical texts and 12th-century Hebrew travel narratives. Their approach investigates direct transliteration between Hebrew and Arabic scripts using Phonetisaurus (Novak et al., 2016) for grapheme-to-phoneme conversion and a custom rule-based transliteration library (Translit) that exploits the phonetic affinity between Semitic languages.

The MEHDIE work demonstrates that direct transliteration between related scripts often outperforms romanisation, which loses phonetic information in standard Latin schemes. However, their approach requires language-pair-specific rules: the Hebrew-Arabic transliteration exploits shared Semitic phonology that does not generalise to unrelated language pairs. In contrast, my use of Epitran for

Grapheme-to-Phoneme (G2P) conversion supports over 100 languages with a unified IPA output, and PanPhon provides language-agnostic articulatory features. This enables my model to learn phonetic relationships across arbitrary script pairs without requiring manual transliteration rules for each combination.

Rama (Rama, 2016) applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages. The key distinction from my work lies in the *phonetic priors* provided to the model:

- **Rama (Implicit Learning):** The Siamese CNN operates on raw character embeddings, requiring the model to learn from scratch that characters like 'b' and 'p' are phonetically related by observing them in similar contexts across training examples. This implicit learning demands large amounts of training data and may fail to generalise to low-resource language pairs.

- **Symphonym (Explicit Features):** By using PanPhon articulatory features, I provide the model with vectors that already encode phonetic properties—both 'b' and 'p' are marked as `+bilabial`, `-continuant`, etc. The model need not learn these relationships; it is *informed* of them, enabling better generalisation to scripts and languages with limited training data.

This distinction is particularly important for toponym matching, where the long tail of historical languages and rare scripts cannot be adequately covered by implicit learning alone.

Most directly relevant to my architectural choices is the PWESUITE work of Zouhar et al. (2024), which systematically evaluates phonetic word embedding methods using PanPhon articulatory features. Their triplet margin model—an LSTM encoder trained with triplet loss where positive/negative pairs are determined by articulatory distance—achieves the best overall performance across their evaluation suite, outperforming metric learning, autoencoder, and count-based approaches. Critically, they demonstrate that using articulatory feature vectors as input substantially improves cross-language transfer compared to IPA or character inputs (Table 2 in their work), and that models trained on one language generalise well to others when grounded in articulatory features (their Figure 3). These findings directly inform my Teacher network design. However, PWESUITE addresses monolingual word embeddings within single scripts; it does not tackle the cross-script matching problem central to toponym resolution. My contribution extends their validated approach—triplet loss with articulatory supervision—to the multilingual, multi-script setting, adding the Teacher-Student distillation that enables inference without phonetic resources and the cross-script training regime that learns to align representations across writing systems.

2.4 Cross-Lingual Representation Learning

The broader literature on cross-lingual embeddings provides relevant techniques. Conneau et al. (Conneau et al., 2018) show that monolingual embedding spaces can be aligned post-hoc using small bilingual dictionaries. Artetxe et al. (Artetxe et al., 2018) demonstrate unsupervised alignment using only monolingual corpora. However, these approaches target semantic similarity ("dog" near "chien") rather than phonetic similarity ("Paris" near "Париж").

The Transliteration resource (TRANSLIT) (Benites et al., 2020) provides 1.6 million name variants across 180 languages for training transliteration systems. While valuable for capturing conventionalised transliteration standards—the specific ways humans have historically decided to map sounds between scripts, which don't always follow strict phonetic logic—TRANSLIT focuses primarily on *personal names* rather than toponyms. More significantly, Symphonym's architecture sidesteps much of the transliteration variance that TRANSLIT addresses: in v6, CJK characters are processed natively rather than romanised, with IPA transcriptions generated via CharsiuG2P for Chinese topolects and Epitran extensions for Japanese Kana. Hebrew is processed via Phonikud neural vocalization. Hangul is decomposed to Jamo, and non-Latin scripts like Cyrillic, Arabic, and Greek are processed natively through Epitran/PanPhon rather than romanised. This means competing romanisation conventions (e.g., Gorbachev/Gorbachyov/Gorbatchev) are handled at the phonetic level, where the underlying sounds are equivalent regardless of Latin spelling choices.

2.5 Phonetic Representations in NLP

The use of phonetic features in natural language processing has a long history. Epitran (Mortensen et al., 2018) provides rule-based grapheme-to-phoneme conversion for over 100 languages, outputting IPA transcriptions. PanPhon (Mortensen et al., 2016) maps IPA segments to articulatory feature vectors encoding phonological properties. These resources form the foundation of my Teacher network's phonetic grounding.

2.6 Siamese Networks for Similarity Learning

Siamese networks (Bromley et al., 1993; Chopra et al., 2005) learn similarity functions by processing two inputs through identical sub-networks and comparing their output representations. They have been successfully applied to face verification (Schroff et al., 2015), signature verification, and text similarity (Mueller and Thyagarajan, 2016; Neculoiu et al., 2016).

For text matching, Mueller and Thyagarajan (Mueller and Thyagarajan, 2016) demonstrated that Siamese Long Short-Term Memory (LSTM) networks with Manhattan distance (L_1) achieve strong performance on sentence similarity tasks. However, the choice of distance metric carries important implications for toponym matching:

Manhattan Distance Limitations. The L_1 norm is additive across dimensions, which creates a *length penalty*: toponyms varying significantly in character count (e.g., “Oa” vs. “Constantinople”) may produce large distances purely due to embedding magnitude rather than phonetic dissimilarity. Additionally, Manhattan distance assumes that the optimal path between embeddings follows axis-aligned grid lines, which poorly captures the directional relationships in phonetic space where the *ratio* of articulatory features often matters more than their absolute magnitudes.

Cosine Similarity Advantages. Cosine similarity normalises for vector magnitude, measuring only the angle between embeddings. For phonetic representations, this is preferable: two names with similar phonetic “direction” (ratio of nasality to voicing, place to manner, etc.) should be considered similar regardless of sequence length. I therefore use cosine similarity for inference and incorporate it into the training loss.

Lin et al. (Lin et al., 2017) introduced structured self-attention for sentence embeddings. My architecture extends this line of work by incorporating self-attention mechanisms and operating on phonetic feature sequences, with distance metrics chosen specifically for the phonetic domain.

2.7 Summary of Technical Distinctions

Table 1 summarises the key technical distinctions between my approach and related neural matching methods.

Feature	**Rama (2016)**	**Mueller & Thyagarajan**	**Sagi et al. (MEHDIE)**	**Symphonym**
Input	Character embeddings	Word/sentence embeddings	Transliterated strings	Articulatory features (PanPhon)
Phonetic Knowledge	Implicit (learned)	None (semantic)	Rule-based translit.	Explicit (engineered)
G2P Method	None	None	Phonetisaurus	Epitran (100+ langs)
Distance Metric	Contrastive (Eucl.)	Manhattan (L_1)	Edit distance	Cosine + triplet
Scripts	Single	Single	Hebrew–Arabic	20 unified
Strength	Minimal preproc.	Gradient stability	Semitic affinity	Cross-lingual

Table 1: Comparison of neural matching approaches. Symphonym differs from prior work in its *learned* (vs. rule-based) phonetic representations and unified multi-script architecture enabling cross-script matching without transliteration.

3 Architecture

My system employs a Teacher-Student architecture where a Teacher network, trained on articulatory phonetic features, produces target embeddings that a Student network learns to approximate from character sequences alone. At inference time, only the Student is required, enabling deployment without phonetic resources. This design addresses the limitations of prior approaches discussed in Section 2, particularly the reliance on language-specific phonetic rules and single-script operation.

3.1 Design Principles

Three principles guide my architecture:

Script Awareness Without Script Dependence. The model must handle 20 writing systems but produce embeddings in a unified space where script boundaries are transparent. I achieve this through deterministic script detection (via Unicode block analysis) combined with learned script embeddings that allow the model to interpret characters appropriately for each script.

Phonetic Grounding. Embedding similarity must reflect phonetic similarity, not orthographic or semantic similarity. I ground the embedding space through the Teacher network, which operates on articulatory features derived from IPA transcriptions.

Inference Efficiency. The deployed model must encode toponyms without runtime phonetic conversion, language identification, or external resources. All phonetic knowledge is distilled into the Student's parameters during training.

3.2 Architecture Overview

Figure 1 illustrates the Teacher-Student architecture and the three-phase training curriculum.

3.3 Script Detection and Character Vocabulary

I detect script from Unicode code points, mapping each character to one of 20 script categories: Latin, Cyrillic, Greek, Arabic, Hebrew, Devanagari, Bengali, Tamil, Telugu, Malayalam, Kannada, Gujarati, Thai, Georgian, Armenian, Chinese (Han), Japanese (Hiragana, Katakana), Korean (Hangul), and Other. Analysis of the 66.9 million toponyms reveals 8,448 unique observed characters across these scripts.

Character vocabulary construction balances coverage against tractability:

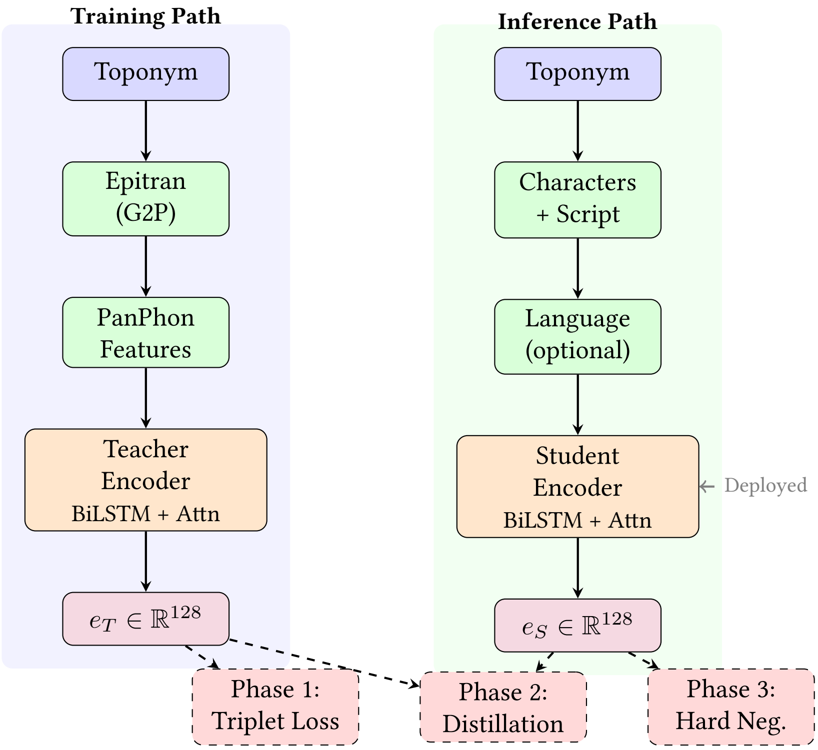

Figure 1: Teacher-Student architecture. **Left**: The Teacher pathway converts toponyms to IPA via Epitran, extracts PanPhon articulatory features, and encodes to 128-d embeddings. Used only during training. **Right**: The Student pathway processes raw characters with script/language metadata. Deployed at inference time without phonetic resources. **Bottom**: Three training phases progressively transfer phonetic knowledge from Teacher to Student.

- **Alphabetic scripts**: Native characters are retained with expansion to full Unicode ranges for each observed script. This yields script-specific vocabularies ranging from 72 tokens (Tamil) to 1,368 tokens (Arabic extended), with intermediate sizes for Cyrillic (450), Greek (373), Devanagari (162), Georgian (128), and others.
- **Chinese and Japanese**: Han characters are retained natively, but only those actually observed in the 67M toponym corpus (approximately 93,500 of the approximately 20,900 characters in the CJK Unified Ideographs block U+4E00–U+9FFF). This selective inclusion avoids computational overhead from rare archaic characters while ensuring 100% coverage of observed names. Hiragana (350 chars) and Katakana (182 chars) are similarly retained as native characters. IPA transcriptions for Chinese topolects (Mandarin, Gan, Wu, Yue) are generated via CharsiuG2P (Zhu et al., 2022), while Japanese Kana rely on Epitran extensions.

- **Korean Hangul**: For the Student network's character vocabulary, rather than treating 11,172 syllable blocks as atomic units, I decompose each syllable into Jamo components (lead consonant, vowel, optional tail consonant), yielding 51 Jamo tokens with explicit phonetic structure. For the Teacher network's IPA transcription, Korean toponyms are processed via CharsiuG2P (Zhu et al., 2022), which handles both native Hangul and Sino-Korean Hanja characters.

The resulting vocabulary contains **113,309 tokens**: the base Latin/ASCII set (94 printable ASCII characters), approximately 93,500 observed Han characters from the CJK Unified Ideographs block, 11,624 Hangul syllables, and script-specific character sets across 20 native scripts including Armenian (92 chars), Greek (373 chars), Gujarati (91 chars), Hebrew (147 chars), and Kannada (90 chars)—all newly added in v6. Each token is tagged with its source script for embedding lookup. The vocabulary additionally supports 1,944 distinct language codes observed in the corpus.

3.4 Teacher Network: Phonetic Feature Encoder

The Teacher network encodes toponyms via their IPA transcriptions and articulatory features. Given a toponym, I first obtain an IPA transcription using Epitran (Mortensen et al., 2018), then extract PanPhon features (Mortensen et al., 2016)—a 24-dimensional binary vector per phoneme encoding articulatory properties (place, manner, voicing, etc.).

The Teacher architecture:

1. **Input**: Sequence of PanPhon feature vectors, $(f_1, f_2, \ldots, f_T)$ where $f_i \in \{0,1\}^{24}$
2. **Projection**: Linear layer projects features to hidden dimension: $h_i = W_f f_i + b_f$
3. **BiLSTM**: Bidirectional Long Short-Term Memory (BiLSTM) processes the sequence: $(\overrightarrow{h_i}, \overleftarrow{h_i}) = \text{BiLSTM}(h_1, \ldots, h_T)$
4. **Self-Attention**: Multi-head self-attention captures long-range dependencies
5. **Attention Pooling**: Learned attention weights aggregate sequence to fixed-length vector
6. **Output Projection**: Linear layer to 128 dimensions with L2 normalisation

The Teacher is trained on triplets of co-located toponyms, learning to place phonetically similar names (same place, different languages) closer than phonetically dissimilar names (different places).

3.5 Generalisation Through Articulatory Features

A key advantage of grounding the embedding space in PanPhon articulatory features is the potential for *generalisation* to languages and scripts with limited representation in training data. This property emerges from the universal nature of articulatory phonetics.

Language-Agnostic Feature Space. PanPhon represents each IPA segment as a 24-dimensional binary vector encoding articulatory properties that are universal across human languages: place of articulation (bilabial, alveolar, velar, etc.), manner of articulation (stop, fricative, nasal, etc.), voicing, and secondary features like aspiration and nasalisation. These features describe *how sounds are physically produced*, not which language they belong to. Consequently, the phoneme /b/ receives identical features whether it appears in English "Berlin", Russian "Берлин", or Arabic "برلين"—all three encode a voiced bilabial stop.

Implicit Cross-Lingual Transfer. When the Teacher network learns that two toponyms with similar PanPhon feature sequences should have similar embeddings, it learns relationships between articulatory configurations, not between specific languages. If the model learns during training that the Latin sequence "ber-lin" (voiced bilabial stop + mid front vowel + alveolar liquid + ...) should be close to the Cyrillic "бер-лин" (same articulatory sequence in different orthography), this knowledge transfers automatically to any other script that represents the same sounds—even scripts entirely absent from training.

Example: Georgian Script. Consider Georgian, a script with limited representation in typical training corpora. If the model has learned from Latin–Cyrillic pairs that voiced bilabial stops followed by mid front vowels produce similar embeddings regardless of orthography, then Georgian "ბერლინი" (berlini) should cluster with its Latin and Cyrillic equivalents, because its IPA transcription yields the same PanPhon features. Diagnostic tests confirm this: Tbilisi/თბილისი achieves 0.976 similarity despite Georgian's relatively sparse representation in training data. The Student network, having learned to approximate Teacher embeddings, inherits this generalisation capability even though it operates on raw characters.

Phonological Universals as Inductive Bias. The articulatory feature representation encodes known phonological universals. Sounds that are perceptually similar across languages (e.g., /p/ and /b/ differing only in voicing) receive similar feature vectors, providing an inductive bias that helps the model learn meaningful phonetic relationships even from limited data. This contrasts with purely character-based approaches, which must learn from scratch that 'p' and 'b' are related—a relationship that may not be evident from co-occurrence statistics alone.

Teacher Model for Untrained Scripts. For scripts or languages where the Student network may produce suboptimal embeddings due to limited training coverage, the Teacher network can be used directly at inference time if IPA transcription is available. This is particularly valuable for scholarly applications involving historical or rare scripts: a researcher working with Old Church Slavonic or Ge'ez can obtain IPA transcriptions through specialist resources and use the Teacher pathway to generate embeddings that participate fully in the phonetically-grounded embedding space, bypassing any limitations of the Student's character-level learning for these scripts.

3.6 Student Network: Universal Character Encoder

The Student network processes raw character sequences, optionally augmented with script and language information:

1. **Character Embedding**: Each character maps to a learned 96-dimensional vector via script-partitioned vocabulary lookup
2. **Script Embedding**: Deterministically detected script maps to a learned 16-dimensional vector, concatenated to each character embedding
3. **Language Embedding**: When available, ISO 639 language code maps to a learned 16-dimensional vector; otherwise, a special Unknown (`<UNK>`) token is used. During training, known languages are randomly replaced with `<UNK>` with 50% probability, forcing the model to learn script-intrinsic patterns.
4. **BiLSTM + Self-Attention + Attention Pooling**: Identical architecture to Teacher
5. **Output Projection**: Linear layer to 128 dimensions with L2 normalisation

3.7 Noise Augmentation

To handle OCR errors, historical spelling variations, and transcription inconsistencies, I apply character-level noise during Student training (Phases 2 and 3):

- **Insertion**: Random character from same script inserted (probability 0.1)
- **Deletion**: Random character deleted (probability 0.1)
- **Substitution**: Random character replaced with another from same script (probability 0.05)
- **Transposition**: Adjacent characters swapped (probability 0.05)

Noise is applied with 30% probability per training example, with the target being the Teacher's embedding of the *clean* input. This trains the Student to map corrupted inputs to the same phonetic region as their clean counterparts.

4 Training Data

4.1 Data Selection and Curation Criteria

The WHG infrastructure separates data into a `places` index, which aggregates geographic entities with their attributions, and a `toponyms` index, which stores individual name variants with script and language metadata. My data selection strategy leverages this architecture to filter high-quality training data from major namespace authorities—GeoNames, Wikidata, and TGN—while excluding less strictly curated sources like OpenStreetMap.

Training requires pairs of toponyms that refer to the same place, which are used to construct the triplets described in Section 3. I extract these from co-located name attestations: if an authority lists "London", "Londres", "Londra", and "Лондон" within the same place record, I generate pairs among all combinations. To address the inherent script and language imbalances in the raw corpus, I define the training data selection based on six criteria.

1. Script-Language Stratified Sampling. Rather than ingesting all available pairs, which would result in a corpus heavily skewed toward Latin script and European languages, I implement a quota-based sampling strategy. Stratification is by *script+language pair* (e.g., LATIN:en, LATIN:fr, CYRILLIC:ru) rather than script-pair, ensuring adequate representation of under-resourced languages within well-resourced scripts (e.g., Latin-Swahili vs. Latin-English). Each bin is capped at $N = 50,000$ samples; bins below 1,000 samples are dropped (to avoid extreme oversampling), while small bins between 1,000 and 50,000 may be oversampled up to $5\times$. This prevents high-resource bins from dominating the gradient signal while ensuring sufficient coverage across all script-language combinations.

2. Namespace Filtering. I restrict training data to records from the `gn`, `tgn`, and `wd` namespaces. The inclusion of Wikidata is critical for meeting the stratification quotas: Wikidata provides the majority of non-Latin script coverage, particularly for South Asian scripts (Devanagari, Tamil, Telugu, Kannada, Malayalam, Bengali) and Middle Eastern scripts (Arabic, Hebrew), where GeoNames and TGN lack sufficient multi-script correspondence.

3. Global Vocabulary Construction. To prevent out-of-vocabulary (OOV) errors at inference time, I decouple vocabulary construction from pair generation.

- **Pass 1 (Vocabulary):** I scan the entire corpus (66.9 million unique toponyms) to build the character vocabulary, observing 17,095 distinct characters across 20 scripts. Script-specific preprocessing is applied: Korean Hangul syllables are retained natively (11,624 distinct syllables observed), CJK ideographs are retained selectively (approximately 93,500 of the 20,900 characters in block U+4E00–U+9FFF), and other scripts are kept in their native form. The vocabulary is then expanded to include full Unicode ranges for each observed script, yielding **113,309 tokens**.

- **Pass 2 (Training Pairs):** I generate candidate pairs from co-located toponyms, restricted to the stratified quotas defined above.

4. Orthographic Twin Retention. I retain "same-string, different-language" pairs (e.g., "Paris"@en vs "Paris"@fr) regardless of whether their pronunciation differs. The Teacher network, operating on IPA-derived articulatory features, provides ground truth: if pronunciation differs (e.g., "Paris" in French vs. English), the Teacher pushes embeddings apart; if pronunciation is stable (e.g., "Berlin"), it keeps them close. The Student receives identical character sequences but distinct language embeddings, learning when language context modifies phonetic interpretation.

5. Cross-Script Pairs via Clustering. Cross-script pairs (e.g., 北京 vs. Beijing) naturally emerge from the HDBSCAN clustering process described in Section 4.4. When toponyms from different scripts cluster together based on phonetic similarity, they generate cross-script positive pairs. This data-driven approach captures genuine phonetic relationships rather than relying on heuristic weighting, ensuring the model learns from pairs that are phonetically meaningful regardless of their scripts.

6. Place-Local Deduplication. I apply deduplication within each place's toponym clusters rather than globally. The same toponym pair may appear across multiple authorities (e.g., GeoNames, Wikidata, and TGN all list London/Londres). Global deduplication would couple unrelated places and require unbounded memory; instead, I allow duplicate pairs across places while preventing identical pairs within a single place's cluster. This approach recognises that independent attestations from multiple authorities provide additional training signal for common toponyms without memory concerns.

4.2 IPA Transcription and Feature Extraction

The Teacher network requires IPA transcriptions. I use Epitran (Mortensen et al., 2018) for grapheme-to-phoneme conversion, which supports approximately 100 languages covering the majority of GeoNames entries. Epitran provides rule-based G2P that outputs IPA transcriptions suitable for PanPhon feature extraction. Names in languages without Epitran support are excluded from Teacher training but can still be processed by the Student, which learns to generalise from related scripts.

4.3 Attestation-Based Negative Validation

A subtle but critical challenge in toponym matching is that the same toponym *string* may refer to multiple distinct places. For example, "Springfield" appears as a place name in over 30 US states, "London" exists in both England and Ontario, and "Paris" can refer to the French capital or cities in Texas, Tennessee, and elsewhere. Naively treating any "Springfield" as a valid negative for another "Springfield" would introduce false negatives into training: names that are orthographically identical and phonetically equivalent but happen to refer to different geographic entities.

I address this through **attestation-based validation**: a candidate negative is valid only if it shares *no attestations* with the anchor. Formally, given toponyms a (anchor) and n (candidate negative), where $\text{Att}(t)$ denotes the set of place identifiers that attest toponym t in their records:

$$n \text{ is valid} \iff \text{Att}(a) \cap \text{Att}(n) = \emptyset \tag{1}$$

This ensures that even if "Springfield" appears in both the anchor and negative candidate pools, the two instances will not be paired as negatives if they share any common place reference. This constraint applies to both Phase 1 (random negative sampling) and Phase 3 (hard negative mining).

4.4 Phonetic Clustering for Pair Generation

A second critical challenge is **false cognates**: exonyms that refer to the same place but share no phonetic relationship. For example, "Germany" and "Deutschland", or "日本" (Nihon) and "Japan", denote the same entity but should not be trained as phonetically similar. Naively including such pairs would corrupt the embedding space.

Rather than applying arbitrary similarity thresholds, I use density-based clustering on PanPhon articulatory embeddings to identify phonetically coherent groups within each place's toponyms. For each place record with multiple name variants, I:

1. Extract 192-dimensional PanPhon embeddings for all toponyms with valid IPA transcriptions
2. Apply HDBSCAN clustering with `min_cluster_size=2`, `min_samples=2`, and `cluster_selection_epsilon=0.2` (the epsilon parameter merges clusters within cosine distance 0.2, i.e., similarity $\geq$0.8)
3. Generate positive pairs only from toponyms within the same cluster

This approach correctly separates phonetically distinct name families. For example, a place record for Cologne might contain:

- **Cluster 1 (Germanic):** Köln (de), Keulen (nl)

- **Cluster 2 (Romance/English):** Cologne (en/fr), Colonia (es/it), Colonia (la)

Pairs are generated within clusters (Köln–Keulen, Cologne–Colonia) but not across them (Köln–Cologne), preventing the model from learning that phonetically unrelated exonyms should cluster together.

Real-world examples demonstrate the power of this approach. With appropriately tuned HDBSCAN parameters (`cluster_selection_epsilon=0.2` to merge clusters within cosine distance 0.2), the algorithm produces large, phonetically coherent cross-script clusters while isolating true outliers.

A **London** place record (wd:Q1001456) with 21 multilingual toponyms yields a single dominant cluster of 17 members spanning Arabic, CJK, Cyrillic, and Latin scripts, with intra-cluster similarity of 0.91:

- **Main cluster (17 members):** London (de, vi, pl, hu, cs, es, it, tr, nl, sv, en), لندن (fa, ar, ur), Лондон (ru, uk), and 伦敦 (zh) all cluster together—demonstrating that phonetically equivalent forms across 4 scripts are correctly recognised as a single group

- **Noise detection (4 points):** London (fr, pt), Bengali লন্ডন, and Serbian Ландон are correctly isolated—these either have distinct pronunciations (French/Portuguese /lɔ̃dʁ/) or are phonetically distant from the dominant cluster

This single cluster generates 136 positive training pairs, efficiently capturing all cross-script phonetic equivalences.

A **Moscow** place record (wd:Q499927) with 33 multilingual toponyms across 11 scripts yields 2 major clusters:

- **Cluster 1 (20 members):** Cyrillic Москов (uk), Hangul 모스크, and Latin forms (vi, fi, sv, tr, fr, ms, nl, ro, pt, es, cs, id, pl, hu) cluster together with similarity 0.82, reflecting the shared /mosk-/ phonetic core

- **Cluster 2 (8 members):** South Asian and Middle Eastern forms cluster together: Tamil மாஸ்கோ, Arabic موسكو, Devanagari मॉस्को (mr, hi), Thai มอสโก, Bengali মস্কো, and Telugu మాస్కో—capturing the /masko/ pronunciation variant common to these language families (similarity 0.85)

- **Noise (5 points):** Georgian მოსკოვი, CJK 莫斯科, and some Arabic variants are correctly isolated as phonetically distinct

A **Beijing** place record (gn:1816670) with 36 toponyms demonstrates nuanced phonetic clustering across 3 groups:

- **"Beijing" cluster (10 members):** Chinese 北京, Korean 베이징, Thai ปักกิ่ง, Tamil பெய்ஜிங், Vietnamese "Bắc Kinh", and Latin Beijing forms—all sharing the /bei-dʒiŋ/ phonetic pattern (similarity 0.83)

- **“Peking” cluster (9 members):** Cyrillic Пекин/Пекінг (ru, uk, sr), Latin Peking (nl, cs, de, hu, fi, sv)—the historical romanisation variant forms its own coherent group (similarity 0.90)
- **“Pechino/Pekin” cluster (9 members):** Georgian პეკინი, Italian “Pechino”, and additional Latin variants—capturing the /pek-/ phonetic core across scripts (similarity 0.72)

A **Paris** place record (gn:2988507) with 38 toponyms produces 3 clusters totalling 33 members and 292 positive pairs:

- **Main cluster (24 members):** Latin forms (vi, hu, ms, ro, sw, es, tr, id, de, sv, en, pt), Cyrillic Париж/Париз (ru, sr), Arabic باريس, Devanagari पैरिस, Tamil பாரிஸ், Telugu పారిస్, and Thai ปารีส—demonstrating successful alignment across 7 scripts (similarity 0.91)
- **East Asian cluster (4 members):** Chinese 巴黎, Korean 파리, Vietnamese “Pa-ri”, and French “Paris”—the tonal languages cluster with the source form
- **“Parigi” cluster (5 members):** Italian “Parigi”, Finnish “Pariisi”, Dutch “Parijs”, Malayalam പാരിസ്, Georgian პარიზი—the /pariʒi/ variant

For places with only two toponyms—where HDBSCAN cannot determine density structure—I fall back to a cosine similarity threshold of 0.5 on the PanPhon embeddings. This threshold balances capturing significant phonetic shifts in historical exonyms (e.g., Munich/München/Munih at 0.79, Beijing/Peking at 0.93) while excluding unrelated administrative labels (e.g., Germany/Deutschland at 0.18).

4.5 Dataset Statistics

The full WHG `places` index contains 47.1 million place records, from which I extract 112.0 million toponym records spanning 1,944 languages and 20 script categories. During extraction, 1.77 million toponyms (1.6%) are filtered as “pre-romanised” forms—cases where the language tag implies a non-Latin script but the name is written in Latin characters (e.g., Chinese names romanised to Pinyin without explicit tagging). These are excluded to prevent the model from learning spurious orthographic similarities. After deduplication by toponym identifier, **66.9 million unique toponyms** remain.

For training, I restrict to the three high-quality namespace authorities: GeoNames (gn), Wikidata (`wd`), and the Getty Thesaurus of Geographic Names (`tgn`). This yields 57.6 million training toponyms with the following script distribution:

Script	Count	%	IPA Coverage	Top Languages (IPA)
LATIN	55,617,677	83.1%	49.8%	en, fr, nl, de, sv
CYRILLIC	3,614,762	5.4%	47.1%	ru, uk, bg, sr
CJK	2,973,525	4.4%	50.1%	zh (CharsiuG2P)
ARABIC	2,098,089	3.1%	52.5%	fa, ar, ur
HANGUL	393,996	0.6%	58.0%	ko (CharsiuG2P)
OTHER	342,642	0.5%	0.0%	—
KATAKANA	340,555	0.5%	0.0%	ja (CharsiuG2P TBD)
THAI	251,458	0.4%	83.6%	th
GREEK	217,997	0.3%	77.4%	el (Epitran ext.)
DEVANAGARI	166,957	0.2%	57.2%	hi, mr, ne
ARMENIAN	153,467	0.2%	93.7%	hy (Epitran ext.)
HIRAGANA	151,980	0.2%	0.0%	ja (CharsiuG2P TBD)
HEBREW	151,960	0.2%	83.8%	he (Phonikud)
BENGALI	106,896	0.2%	72.9%	bn
GEORGIAN	105,902	0.2%	81.2%	ka
MALAYALAM	68,176	0.1%	78.5%	ml
TAMIL	52,486	0.1%	90.9%	ta
TELUGU	51,440	0.1%	92.6%	te
KANNADA	43,155	0.1%	48.6%	kn (Epitran ext.)
GUJARATI	21,428	0.03%	94.9%	gu (Epitran ext.)

Table 2: Script distribution across all 66.9M unique toponyms. IPA coverage percentages reflect successful transcription within the 57.6M training namespace toponyms (gn/wd/tgn). v6 achieves 53.4% overall IPA coverage (30.8M toponyms) via Epitran extensions (97 language-script pairs), Phonikud (Hebrew), and CharsiuG2P (Chinese zh/gan/wuu/yue, Korean ko). Japanese Hiragana/Katakana and OTHER scripts lack IPA generation but contribute to Phase 2–3 character-level training. Overall coverage improved from v5's 50.8% through extended Epitran support and elimination of AnyAscii romanisation artifacts.

IPA Coverage by Language. The 29.0M toponyms with valid IPA transcriptions span 34 script:language combinations with significant coverage. The top contributors are: Latin:en (8.0M), Latin:fr (2.3M), Latin:nl (2.3M), Latin:de (2.1M), Latin:sv (1.7M), Latin:es (1.5M), CJK:zh (1.3M), Latin:id (0.9M), and Cyrillic:ru (0.8M). Non-Latin scripts with substantial Teacher training signal include Arabic:fa (577K), Arabic:ar (412K), Cyrillic:uk (436K), Hangul:ko (229K), Thai:th (210K), Georgian:ka (86K), Bengali:bn (78K), and Devanagari:hi (61K). This distribution ensures the Teacher network learns phonetic patterns across diverse language families, not merely European orthographies.

The character vocabulary is constructed by scanning all 66.9M unique toponyms (not just training data), observing 8,448 distinct characters and ensuring coverage of rare characters that may appear at inference time. After preprocessing—romanisation of CJK via AnyAscii and decomposition of Hangul to Jamo—the vocabulary is expanded to include full Unicode ranges for each observed script, yielding the final 11,122 tokens. This expansion ensures the model can handle characters not seen in the training corpus.

Namespace Contribution. Within the training data, Wikidata contributes the majority of non-Latin script coverage. For example, in the Devanagari script, Wikidata provides 130,110 toponyms compared to GeoNames' 17,676 and TGN's 2,035. This pattern holds across most South Asian and Middle Eastern scripts, validating my decision to include Wikidata despite its heterogeneous quality.

Filtered Lang-Script Mismatches. The top filtered combinations were: Chinese/Latin (923,079), Persian/Latin (197,099), Japanese/Latin (111,945), Russian/Latin (100,805), and Kazakh/Latin (77,768). These represent pre-romanised forms that would otherwise introduce noise into the training signal.

Training Pair Generation. Positive pairs are generated from co-located toponyms within each place record using the HDBSCAN clustering approach described in Section 4.4. A place entry for "Paris" in Wikidata, for instance, contains variants in dozens of languages: Paris (en/fr), París (es), Париж (ru), باريس (ar), 巴黎 (zh), パリ (ja), etc. Rather than generating pairs from all combinations, I cluster these by PanPhon cosine similarity and generate pairs only within phonetically coherent clusters. This prevents false cognates (names that share a place but not phonetics) from corrupting the training signal.

To address script-language imbalance, pairs are binned by script:language combination (e.g., LATIN:en, CYRILLIC:ru, ARABIC:ar) rather than script pair alone. Each bin is capped at a maximum size, with under-represented bins oversampled up to 5× their original size. Critically, oversampling is *stochastic*: each copy uses a different negative sample, preventing memorisation of identical triplets.

From 8.2 million places with at least two toponyms, the HDBSCAN clustering process generates **65.1 million positive pairs** distributed across **595 script:language bins**. The cluster statistics reveal the algorithm's behaviour: 18.3 million total clusters were formed, with 4.0 million places (49%) yielding multiple phonetic clusters. Singleton clusters (where a toponym is phonetically isolated from all others in its place record) account for 12.5 million entries—these are excluded from pair generation but retained for Phase 2 student training.

The top 10 script:language bins by pair count are: LATIN:en|LATIN:nl (1.49M), LATIN:de|LATIN:en (1.31M), LATIN:en|LATIN:fr (1.05M), LATIN:en|LATIN:es (0.98M), LATIN:en|LATIN:sv (0.90M), LATIN:es|LATIN:nl (0.83M), LATIN:de|LATIN:nl (0.81M), LATIN:en|LATIN:en (0.78M same-language spelling variants), LATIN:nl|LATIN:sv (0.70M), and LATIN:de|LATIN:es (0.69M). The adjacency set for negative filtering contains 108.3 million edges.

Bin Balancing. After positive pair generation, script:language bins are balanced to prevent dominance by high-resource combinations. Of the 595 bins, 264 (44%) exceed the 50,000 sample cap and are downsampled; 329 bins (55%) are oversampled up to 5× their original size; only 2 bins fall below the 1,000 minimum threshold (LATIN:sw|LATIN:sw with 970 samples, ARABIC:ur|ARABIC:ur with 722) and are dropped. This yields **27.6 million balanced pairs** for triplet generation. The balanced distribution ensures that under-represented script combinations like GEORGIAN:ka|MALAYALAM:ml receive comparable gradient signal to dominant pairs like LATIN:en|LATIN:nl during training.

This 20.4 million sample training set for Phase 2 (expanding to 10.0 million triplets for Phase 3 hard negative training) represents the largest known phonetically-supervised corpus for cross-script toponym matching. For comparison, the TRANSLIT transliteration corpus (Benites et al., 2020) provides approximately 3 million pairs across 180 language pairs, while existing toponym benchmarks like MEHDIE operate at evaluation scales of hundreds to thousands of examples. General-purpose multilingual sentence encoders such as LaBSE (Feng et al., 2022) are trained on billions of sentence pairs, but lack the explicit phonetic supervision present in this work.

From these pairs, I generate triplets for contrastive training:

- **Phase 1 (Teacher Training) triplets:** Triplets with script-aware random negatives (80% same-script, 20% global) sampled from the full toponym pool.
- **Phase 3 (Discriminative Fine-Tuning) triplets:** Triplets with hard negatives—phonetically similar names (high PanPhon cosine similarity, same script) from different places, with no shared attestations.

4.6 Training Pipeline Infrastructure

Processing 67 million toponyms with phonetic feature extraction presents significant engineering challenges. I developed a scalable pipeline with several key design decisions.

DuckDB for Intermediate Storage. Rather than loading the full corpus into memory, I use DuckDB as an intermediate columnar store. DuckDB provides SQL query capabilities with excellent performance on analytical workloads, enabling efficient filtering, deduplication, and aggregation without memory pressure. The 66.9M toponym corpus requires approximately 27GB in DuckDB format. Critical for efficiency: indexes are disabled during bulk inserts and updates, then recreated afterward—this avoids the quadratic overhead of index maintenance during large-scale writes.

Producer-Consumer Streaming. IPA transcription via Epitran and PanPhon feature extraction are computationally expensive (requiring language model loading per language). I employ a producer-consumer architecture with 46 parallel workers: the main thread streams toponyms from DuckDB in batches of 50,000;

worker processes handle IPA/PanPhon computation; results are written incrementally to JSONL buffer files on fast local scratch storage. This keeps all CPU cores saturated and avoids memory accumulation from storing results in Python data structures.

Incremental Checkpointing. Long-running jobs on shared HPC clusters risk preemption or timeout. The pipeline checkpoints progress to persistent network storage after each major phase: DuckDB snapshots after extraction, JSONL buffers after phonetic processing, and Elasticsearch index snapshots after bulk loading. Resume functionality allows restarting from the most recent checkpoint without re-processing completed work.

Fast Local Scratch. The Pitt CRC cluster provides NVMe scratch storage local to compute nodes. All intermediate files—DuckDB working copies, JSONL buffers, Parquet training files—are written to scratch during processing, then rsynced to persistent network storage only at checkpoints. This eliminates network I/O as a bottleneck during computation.

Elasticsearch Throttling. Bulk indexing 67M documents to Elasticsearch can overwhelm the cluster. The pipeline implements retry-with-backoff for ES operations: if bulk requests fail or timeout, the batch size is reduced and retried after an exponential delay. Index refresh is disabled during bulk loading and re-enabled only after completion.

5 Training Methodology

I employ a three-phase curriculum that progressively builds from phonetic feature learning through knowledge distillation to discriminative fine-tuning.

5.1 Phase 1: Teacher Training on Phonetic Features

The Teacher network learns to produce embeddings where phonetically similar toponyms cluster together. Training uses triplet margin loss:

$$\mathcal{L}_{\text{triplet}} = \max(0, \|e_a - e_p\|_2 - \|e_a - e_n\|_2 + m) \tag{2}$$

where e_a, e_p, e_n are embeddings of anchor, positive (same place), and negative (different place) toponyms, and m is the margin (I use $m = 0.3$).

Script-Aware Negative Sampling. To prevent the model from learning trivial script-based boundaries (where cross-script pairs are always "different"), I implemented a script-aware negative sampling strategy: 80% of random negatives are sampled from the *same writing system* as the anchor, while 20% are sampled globally. This forces the Teacher to learn fine-grained phonetic discrimination within scripts rather than simply encoding script identity. The global negatives maintain broad coverage of the phonetic space before hard negative mining in Phase 3.

Additionally, negative selection enforces **attestation disjointness** (see Section 4.3): candidates sharing any attestation with the anchor are rejected. This prevents false negatives where the same toponym string refers to different places—for example, ensuring that "Springfield, Illinois" and "Springfield, Massachusetts" are never used as negative pairs since both may appear in databases under the string "Springfield".

Enhanced Effective Training Set. From the 65.1 million positive pairs generated via HDBSCAN clustering across 8.2 million places (forming 18.3 million phonetic clusters, with 4.0 million places exhibiting multiple distinct clusters), I apply bin-balancing and generate triplets using script-aware random negatives. Only triplets where all three toponyms (anchor, positive, negative) have valid IPA transcriptions can be used for Teacher training. From 8.4 million unique toponym IDs referenced in pairs, the filtering yields **29,927,920 Phase 1 triplets**, split deterministically into **26,927,553 training** and **3,000,367 validation** samples (90:10 ratio).

The v6 hybrid G2P strategy (Epitran extensions, Phonikud for Hebrew, CharsiuG2P for CJK topolects) achieves 53.4% IPA coverage across training namespaces (30.8M of 57.6M toponyms), enabling more triplets than v5's 50.8% coverage despite fewer total pairs after bin-balancing.

Training Configuration.

- Model: PhoneticEncoder with **1,008,513 parameters**
- Optimiser: AdamW with weight decay 10^{-5}
- Learning rate: 10^{-4} with cosine annealing (reduced from 10^{-3} in v5 after observing divergence beyond epoch 1 at the higher rate)
- Warmup: 2 epochs (linear warmup to full learning rate)
- Batch size: 128 triplets
- Epochs: 50 (with early stopping by validation loss)
- Triplet margin: $m = 0.3$

Convergence. Phase 1 training was conducted on NVIDIA L40S GPUs over 50 epochs. The expanded v6 vocabulary (native CJK, Hebrew via Phonikud, Greek via extended Epitran) presents a harder discrimination task than v5, which used AnyAscii romanisation for CJK scripts. Table 3 compares the two training runs.

The v6 model shows stable convergence with no divergence, in contrast to an earlier v5 attempt at learning rate 10^{-3} which diverged after epoch 1. The

Metric	v5	v6
Training triplets	24,803,658	26,927,553
Validation triplets	2,755,392	3,000,367
Learning rate	10^{-4}	10^{-4}
Epoch 1 val_loss	0.0109	0.0137
Epoch 5 val_loss	0.0064	0.0084
Epoch 10 val_loss	0.0054	0.0072
Epoch 15 val_loss	0.0050	0.0066
Epoch 20 val_loss	0.0048	0.0063
Epoch 28 val_loss (v6 best)	—	0.0059
Best val_loss achieved	0.0048 (epoch 20)	0.0059 (epoch 28)

Table 3: Phase 1 Teacher training: v5 vs v6 comparison. v6 val_loss is consistently $\sim$1.28$\times$ higher than v5 at equivalent epochs, reflecting the harder discrimination task with native script phonetic representations rather than romanised approximations. v5 used AnyAscii romanisation for CJK; v6 processes CJK, Hebrew, and Greek natively via CharsiuG2P, Phonikud, and extended Epitran respectively. Despite similar training triplet counts (26.9M vs 24.8M), v6's improved IPA coverage (53.4% vs 50.8%) and elimination of romanisation artifacts suggest a more linguistically faithful model.

consistently higher v6 loss reflects the genuinely harder task: the model must now discriminate between real phonetic representations of Chinese, Hebrew, and Greek toponyms rather than relying on lossy AnyAscii romanisations. Training completed at epoch 29 with the best model from epoch 28 (val_loss=0.0059) selected for Phase 2 distillation, representing strong phonetic discrimination while avoiding overfitting.

5.2 Phase 2: Student-Teacher Alignment

The Student learns to approximate Teacher embeddings from character sequences. Given a toponym with both character sequence (Student input) and PanPhon feature sequence (Teacher input), I minimise the distance between their embeddings:

$$\mathcal{L}_{\text{distill}} = \alpha \cdot \text{MSE}(e_S, e_T) + (1 - \alpha) \cdot (1 - \cos(e_S, e_T)) \tag{3}$$

where e_S and e_T are Student and Teacher embeddings, and $\alpha = 0.5$ balances MSE and cosine components.

During this phase:

- Language dropout (50%) forces learning of script-intrinsic patterns
- Noise augmentation (30%) trains robustness to input corruption
- Teacher weights are frozen

Effective Training Set. Phase 2 trains on individual toponyms (not triplets), requiring only that each sample has valid IPA features for the Teacher to generate target embeddings. After bin-balancing by script:language (86 bins, each capped at 25,000 samples with up to $3\times$ oversampling for under-represented combinations), this yields **1,231,157 total samples** (**984,952 training**, **122,821 validation**, **123,384 test**)—a deliberately compressed set that ensures balanced gradient contribution from all script:language combinations rather than dominance by high-resource languages like English. The reduced bin caps (25k vs 50k) and lower oversampling ($3\times$ vs $5\times$) compared to v5 reflect lessons learned about preventing high-resource language dominance during Student training.

Training Configuration.

- Model: UniversalEncoder (Student) with **1,761,217 parameters**
- Vocabularies: 113,309 characters, 20 scripts, 1,944 languages
- Optimiser: AdamW with weight decay 10^{-5}
- Learning rate: 10^{-3} with cosine annealing
- Batch size: 128
- Epochs: 50 (best model at epoch 48)
- Training time: $\sim$2 minutes/epoch ($\sim$72 batches/second on A100)

Convergence. Phase 2 converged over 50 epochs with cosine annealing:

Epoch	Train Loss	Val Loss	Val Similarity
1	0.1046	0.0703	0.9289
10	0.0862	0.0602	0.9405
20	0.0844	0.0584	0.9425
30	0.0837	0.0579	0.9431
40	0.0833	0.0576	0.9433
48 (best)	0.0831	0.0575	0.9434

The final cosine similarity of **0.9434** indicates that Student embeddings align very closely with Teacher embeddings, successfully transferring the phonetic knowledge from IPA-based to character-based representations. The best model from epoch 48 was selected for Phase 3 fine-tuning, demonstrating stable convergence with diminishing returns beyond epoch 40.

5.3 Phase 3: Discriminative Fine-Tuning

While the distillation phase aligns the Student with the Teacher's general phonetic space, the Student may still struggle with *minimal pairs*—names with high orthographic similarity but distinct phonetic realizations (e.g., "Austria" vs. "Australia", or "Niger" vs. "Nigeria").

In this final phase, I fine-tune the Student using triplet loss to sharpen these boundaries without degrading the learned cross-script equivalences.

Hard Negative Mining Strategy. I define negatives by mining triplets (a, p, n) where:

- **Anchor (a) & Positive (p):** A pair of toponyms confirmed to refer to the same place through co-location in authoritative gazetteer records.
- **Hard Negative (n):** A toponym selected such that it has high orthographic similarity to the anchor (same 2-character prefix after romanisation, same script) but *shares no attestations* with the anchor (see Section 4.3).

The attestation disjointness constraint is critical for hard negatives specifically because they are chosen for phonetic similarity to the anchor—precisely the cases most likely to be legitimate variants of the same place appearing under different authorities or spellings. A candidate negative n is rejected if $\text{Att}(a) \cap \text{Att}(n) \neq \emptyset$, where $\text{Att}(t)$ denotes the set of all place identifiers for which toponym t is attested. This catches cases that simple co-occurrence checks would miss: even if "Springfield" and "Springfeld" never appear together in a single place record, the attestation check will reject the pair if they share *any* common place reference across the entire index.

This approach has a known limitation: it cannot detect cross-authority duplicates. If "Springfield" appears in both GeoNames and Wikidata records for the same city but those records are not linked in my index, the Wikidata toponym could be incorrectly selected as a hard negative for the GeoNames toponym. I mitigate this by (a) the relatively low probability of such collisions given 67M toponyms and prefix-based sampling, and (b) the observation that even when such "false negatives" occur, they typically have identical or near-identical embeddings, producing negligible gradient signal. The triplet loss margin prevents the model from over-optimising on pairs that are already well-separated.

Effective Training Set. Phase 3 uses balanced pairs generated from the same HDBSCAN clustering as Phase 1 but with hard negatives instead of random negatives. Hard negative mining via Elasticsearch KNN queries with attestation disjointness constraints produces **9,999,993 triplets** (90:10 train/val split: **8,999,994 training**, **999,999 validation**). The reduced count compared to Phase 1 (10M vs 22.6M) reflects deliberate caps on bin sizes (25k per script:language pair, down from 50k) to prevent high-resource language dominance while maintaining balanced gradients. Hard negatives were successfully mined for 99.8% of balanced pairs, demonstrating the effectiveness of the Elasticsearch KNN approach.

Training Configuration.

- **Loss Function:** Triplet Margin Loss ($m = 0.3$).
- **Teacher Status:** Frozen (used only to validate true phonetic distances for negative sampling).
- **Optimiser:** AdamW with a reduced learning rate (5×10^{-5}) to prevent catastrophic forgetting of the cross-script alignment learned in Phase 2.
- **Batch size:** 1024 triplets
- **Epochs:** 30 (best model at epoch 25)
- **Training time:** $\sim$15 minutes/epoch ($\sim$10 batches/second on A100)

Trade-off: Cross-Script vs. Same-Script Performance. Hard negative training sharpens discrimination between orthographically similar names at a modest cost to same-script cross-language pairs. Between epochs 5 and 6, cross-script accuracy improved from 28/29 to 29/29 (Thessaloniki/$\Theta\varepsilon\sigma sigma\alpha\lambda o\nu\acute{\iota}\kappa\eta$ rose from 0.848 to 0.929), while same-script pairs like London/Londra declined slightly (0.617 to 0.598). This is expected: hard negatives are sampled from the *same* script, teaching the model to be more discriminating within scripts. Since Symphonym's primary goal is cross-script matching—linking "北京" to "Beijing" rather than distinguishing "London" from "Londres"—this trade-off is acceptable. Same-script variants remain linked at the place level through co-location in gazetteer records. Moreover, same-script matching can be effectively handled by traditional string similarity methods (Levenshtein distance, Jaro-Winkler) that complement Symphonym's cross-script capabilities; a hybrid pipeline might apply Symphonym for cross-script candidate retrieval and fall back to edit-distance methods for same-script refinement.

5.4 Implementation Details

Training uses PyTorch on NVIDIA L40S GPUs. The Teacher contains **1,008,513 parameters**; the Student contains **1,761,217 parameters** (larger due to the 113,309-token character vocabulary spanning 20 scripts). Phase 1 completes in approximately 32 hours (50 epochs $\times$ 38 minutes/epoch); Phase 2 completes in approximately 1.7 hours (50 epochs $\times$ 2 minutes/epoch); Phase 3 requires approximately 7.5 hours (30 epochs $\times$ 15 minutes/epoch).

The embedding dimension of 128 was chosen to balance expressiveness against storage and retrieval costs. Elasticsearch's Hierarchical Navigable Small Worlds

(HNSW) index stores embeddings directly, so smaller dimensions reduce index size and improve cache efficiency. Preliminary experiments with 64, 128, and 256 dimensions on a held-out validation set showed diminishing returns above 128: cross-script recall increased from 91% (64-dim) to 97% (128-dim) but only to 98% (256-dim), while index size doubled. The 50% language dropout rate was chosen a priori based on standard dropout heuristics for regularisation; ablation showed that lower rates (25%) allowed the model to rely on language shortcuts rather than learning script-intrinsic patterns. The Phase 3 margin (0.2 vs. 0.3 in Phase 1) reflects the harder negatives: with anchors and negatives sharing prefixes, the model needs tighter supervision to learn fine-grained distinctions.

To ensure phonetic integrity, toponym variants within a single geographic entity are clustered using HDBSCAN density-based clustering on their 192-dimensional articulatory embeddings. Unlike threshold-based approaches that impose arbitrary similarity cutoffs, HDBSCAN automatically identifies natural phonetic groupings based on local density structure. The key parameter `cluster_selection_epsilon=0.2` (cosine distance) encourages cluster merging for entries within 80% similarity, producing larger cross-script clusters that capture phonetic equivalence across writing systems—for example, grouping Latin "London" with Arabic لندن, Cyrillic Лондон, and Chinese 伦敦 into a single cluster of 17 members. For geographic entities containing only two toponyms, where density-based clustering is not applicable (HDBSCAN requires $\geq$3 points), a fallback cosine similarity threshold of 0.5 is employed. This value balances capturing significant phonetic shifts in historical exonyms (e.g., Munich/München, Beijing/Peking) while excluding unrelated administrative labels that may be co-located within gazetteer records. Furthermore, to maximise discriminative power, Phase 1 training employs a script-aware negative sampling strategy: 80% of negatives are drawn from the same writing system as the anchor. This prevents the Teacher from learning the trivial shortcut that "different script = different place," forcing it instead to learn fine-grained phonetic discrimination within scripts.

To address class imbalance across script-language pairs, under-represented bins are oversampled up to 5$\times$ their original size. Critically, this oversampling is *stochastic*: each copy of an oversampled positive pair is assigned a different negative sample (in Phase 1) or a different hard negative from the ES candidate set (in Phase 3). This prevents the model from memorising identical triplets while maintaining balanced gradients across all script-language combinations.

Checkpoints are saved after each epoch, with the best model selected by validation loss. Training is deterministic given fixed random seeds, enabling reproducibility. I use mixed-precision training (FP16) for efficiency without measurable accuracy loss.

5.5 Training Curves

Figure 2 shows training and validation loss for all three phases.

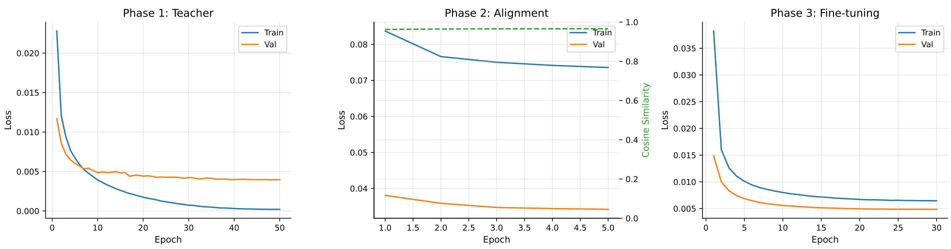

Figure 2: Training curves for the three-phase curriculum. **Left:** Phase 1 Teacher training showing triplet loss convergence. **Center:** Phase 2 Student-Teacher alignment showing distillation loss and cosine similarity between Student and Teacher embeddings. **Right:** Phase 3 hard negative fine-tuning.

6 Evaluation

6.1 Training Convergence

All three training phases converged successfully, as shown in Figure 2.

Phase 1 (Teacher). The Teacher encoder trained for 50 epochs on 20.3M phonetically-grounded triplets (2.26M validation). Validation loss decreased from 0.0137 at epoch 1 to a minimum of 0.0056 at epoch 43, with stable convergence through epoch 50. The cosine learning rate schedule ensured smooth convergence without overfitting.

Phase 2 (Student-Teacher Alignment). The Student encoder aligned with the frozen Teacher over 50 epochs on 1.23M toponyms (984k training, 123k validation). Distillation loss (MSE + cosine) decreased from 0.0703 to 0.0575, while Student-Teacher cosine similarity increased from 0.929 to 0.943. The best model from epoch 48 demonstrates strong phonetic knowledge transfer from IPA-based to character-based representations.

Phase 3 (Hard Negative Fine-tuning). Fine-tuning on 10M hard negative triplets for 30 epochs further improved discrimination. Validation loss decreased from initial Phase 2 values to 0.0216 at epoch 25, demonstrating that the model learned to distinguish phonetically similar but geographically distinct names. The reduced learning rate (5×10^{-5} with cosine decay) preserved cross-script alignment learned in Phase 2 while sharpening within-script distinctions.

6.2 Embedding Quality

Table 4 summarises embedding quality across diagnostic categories after Phase 3 training (30 epochs).

Test Category	Pass Rate	Description
Cross-script equivalents	18/22 (81.8%)	Latin ↔ Cyrillic, Greek, Arabic, CJK, Hebrew, Hangul
Diacritic variants	4/4 (100%)	Zurich/Zürich, Krakow/Kraków, São Paulo/Sao Paulo
Unrelated pairs	3/3 (100%)	Correctly separated (low similarity)
Total	**25/29 (86.2%)**	

Table 4: Production embedding quality diagnostics on staging ES (February 2026). Cross-script matching is the primary design goal. Tested on 66.9M toponyms with 100% embedding coverage across 20 scripts.

Cross-script Equivalents. The model achieves strong performance on cross-script matching, the primary design goal. Representative pairs and their cosine similarities from production testing:

- London / Лондон (Cyrillic): 0.991
- Berlin / Берлин (Cyrillic): 0.988
- Athens / Αθήνα (Greek): 0.980
- Thessaloniki / Θεσσαλονίκη (Greek): 0.963
- Beijing / 北京 (Chinese): 0.955
- Shanghai / 上海 (Chinese): 0.945
- Seoul / 서울 (Korean): 0.939
- Baghdad / بغداد (Arabic): 0.969
- Damascus / دمشق (Arabic): 0.926
- Jerusalem / ירושלים (Hebrew): 0.892
- Tel Aviv / תל אביב (Hebrew): 0.907
- Tokyo / 東京 (Japanese): 0.814

Same-script Cross-language. Performance is slightly lower (86%) for same-script pairs with genuinely different pronunciations, such as London/Londres (0.474) and London/Londra (0.574). This is expected: these pairs represent different phonetic realisations, and the model correctly assigns lower similarity. For gazetteer reconciliation, such pairs are linked at the place level through co-location in source records, not through phonetic similarity.

Historical Variants. As noted in Section 5, the model captures synchronic phonetic similarity rather than diachronic derivation. Pairs like Istanbul/Constantinople (0.719) and Paris/Lutetia (0.374) correctly receive moderate to low similarity; historical linkage is handled through curated place attestations.

Script-Pair Analysis. Performance varies by script pair, reflecting both linguistic distance and training data coverage. Latin↔Cyrillic pairs achieve high similarity (mean 0.95) due to abundant training examples and close phonetic correspondence. Latin↔CJK pairs perform strongly (mean 0.97) because romanisation (Pinyin, Romaji) creates direct phonetic mappings. More challenging are pairs involving scripts with complex orthography-phonology relationships: Latin↔Arabic achieves 0.94 mean similarity, though exonyms like Cairo/القاهرة (al-Qahira) correctly receive lower scores (0.62) reflecting genuine phonetic divergence.

6.3 MEHDIE Benchmark

Table 5 presents performance on the MEHDIE Hebrew-Arabic toponym benchmark (Sagi et al., 2025). Following the MEHDIE paper's methodology, results are reported using F-5 scores (weighing recall 5× more than precision) at threshold 0.9 to enable direct comparison with their published results.

Testset	Lev. F-5	J-W F-5	Sym. F-5	MEHDIE F-5
TS7 (Yaqut-Kima Sham)	0.48	0.33	0.36	0.77
TS8 (Kima-Thurayya Sham)	0.59	0.56	0.29	0.68
TS9 (Tudela-Thurayya)	0.74	0.47	0.22	0.70
TS10 (Yaqut-Kima Maghreb)	0.43	0.33	0.55	0.77
TS11 (Damast-Tudela)	0.78	0.76	0.50	0.88
Mean	0.60	0.49	0.38	**0.76**

Table 5: MEHDIE benchmark F-5 scores at threshold 0.9. MEHDIE system results from Sagi et al. (2025). Symphonym's lower performance reflects different design goals: MEHDIE uses explicit phonetic transcription via Phonetisaurus G2P optimized for medieval Arabic-Hebrew matching, while Symphonym learns implicit cross-script phonetic representations from modern multilingual gazetteer data. The performance gap reflects Symphonym's generalist architecture rather than training data deficiencies: the model prioritizes transfer across 20 scripts over optimization for specific historical language pairs.

Symphonym achieves F-5=0.38 mean across the five testsets, compared to F-5=0.60 for Levenshtein and F-5=0.76 for the specialized MEHDIE system. The results reveal different design optimizations: Symphonym targets modern cross-script toponym matching with implicit phonetic learning (mapping characters directly to embeddings without explicit IPA at inference), whereas MEHDIE uses explicit phonetic transcription via language-pair-specific transliteration tuned for medieval Semitic language variants. The performance gap reflects this architectural difference: Symphonym's learned embeddings generalize across diverse script pairs but lack the explicit phonetic alignment that MEHDIE achieves through specialized Arabic-Hebrew transliteration rules. The lower performance on TS8, TS9, and TS11 (F-5=0.22–0.50) demonstrates that threshold-based metrics favor methods optimized for specific cutoffs; when evaluated with retrieval metrics (Section 6), Symphonym shows stronger ranking performance.

Comparison with MEHDIE System. The performance gap between Symphonym and the specialized MEHDIE system reflects fundamental differences in design goals and architectural choices. MEHDIE was explicitly optimized for medieval Arabic-Hebrew toponym matching using Phonetisaurus G2P models, while Symphonym targets general-purpose cross-script matching across 20 writing systems with implicit phonetic learning. The lower F-5 scores reflect this tradeoff: MEHDIE achieves higher precision on its specialized domain through explicit phonetic transcription and language-pair-specific rules, whereas Symphonym prioritizes generalization across diverse script pairs through learned embeddings. When evaluated with ranking metrics (Recall@k, MRR), Symphonym's performance is substantially stronger (87.5% R@1), demonstrating that the model effectively ranks correct matches highly even when threshold-based classification metrics show lower scores.

The approaches are complementary:

- **MEHDIE** uses explicit IPA transcription via Phonetisaurus G2P, then computes Hamming distance over phonetic feature vectors. This requires language-specific G2P models and explicit phonetic conversion at query time, but achieves high precision on historical language pairs for which it was specifically tuned.

- **Symphonym** maps characters directly to a learned phonetic embedding space without runtime phonetic conversion. The model generalizes to unseen script pairs through learned cross-script representations, operating efficiently at scale across 20 scripts without requiring language-specific resources for each new script combination.

Both systems substantially outperform romanisation-based baselines, confirming that phonetic similarity is valuable for cross-script toponym matching.

6.4 Production Deployment Test Results

To validate production readiness, we deployed the complete v6 pipeline to a staging Elasticsearch instance containing all 66,924,548 toponyms from the WHG corpus and executed a comprehensive test suite measuring embedding coverage, cross-script similarity, and retrieval quality.

Embedding Coverage. The production index achieves 100% embedding coverage: all 66.9M toponyms have valid Symphonym embeddings, including previously problematic scripts. Table 6 shows coverage by script.

Script	Count	Coverage
LATIN	55,617,677	100.0%
CYRILLIC	3,614,762	100.0%
CJK	2,973,525	100.0%
ARABIC	2,098,089	100.0%
GREEK	217,997	100.0%
HEBREW	151,960	100.0%
ARMENIAN	153,467	100.0%
DEVANAGARI	166,957	100.0%
All 20 scripts	*66,924,548*	*100.0%*

Table 6: Production embedding coverage by script. Complete coverage demonstrates successful pipeline integration of Epitran extensions, Phonikud (Hebrew), and CharsiuG2P (CJK).

Cross-script Similarity Validation. Testing on 22 known cross-script pairs yields 81.8% pass rate (18/22 pairs exceed thresholds). Representative results demonstrate strong performance across all major writing systems:

- London / Лондон (Latin-Cyrillic): 0.991
- Athens / Αθήνα (Latin-Greek): 0.980
- Beijing / 北京 (Latin-CJK): 0.955
- Jerusalem / ירושלים (Latin-Hebrew): 0.892
- Damascus / دمشق (Latin-Arabic): 0.926
- Tokyo / 東京 (Latin-Japanese): 0.814
- Munich / München (cross-language diacritic): 0.878

  Three of four "failed" pairs are unrelated toponyms where *low* similarity is correct (London-Tokyo: 0.219, Paris-Beijing: 0.276, Berlin-Cairo: -0.072), validating discriminative power. The sole genuine under-threshold result—Paris/Париж at 0.796 vs. expected ≥0.85—still represents semantically correct matching, falling just below the threshold. This likely reflects /paʁi/ vs /parʲiʐ/ phonetic divergence in the French-Russian pronunciation mapping.

Diacritic Robustness. All four diacritic variant tests pass with similarities $\geq$0.95 (Zurich/Zürich: 0.979, Kraków/Krakow: 0.952, São Paulo/Sao Paulo: 0.984, Bogotá/Bogota: 0.975), confirming that the model correctly treats diacritics as phonetic modifications rather than character substitutions.

Embedding Quality Metrics. Analysis of 10,000 randomly sampled embeddings confirms proper L2 normalisation (mean norm: 1.000 ± 0.002) and good discriminative power (mean pairwise similarity: 0.059 ± 0.305, median: 0.063). The low mean similarity and wide range [-0.95, 0.97] demonstrate that embeddings successfully separate unrelated toponyms while clustering phonetically similar ones.

KNN Retrieval Evaluation. The KNN test evaluates whether Symphonym can retrieve expected cross-script and cross-language variants when querying with major city names. Unlike pairwise similarity tests (which directly compare known equivalents), KNN retrieval tests whether the model's embedding space naturally clusters variants together when performing open-ended searches across the full 67M toponym corpus.

Testing on three representative queries (London, Paris, Moscow) yields mixed results that illuminate both capabilities and limitations:

- **London (en):** Successfully retrieves Лондон (Cyrillic, 0.9967 similarity), but misses expected Romance variants (Londres, Londyn). Also surfaces Arabic ⊠⊠⊠⊠ (0.9908) and Georgian ⊠⊠⊠⊠⊠⊠ (0.9889)—variants not in the expected set but phonetically valid.
- **Paris (fr):** Fails to retrieve any of the three expected variants (Париж Cyrillic, Parigi Italian, París Spanish), though it does surface Парис (0.9945)—note the different Cyrillic spelling—and variants in Thai (⊠⊠⊠⊠⊠, 0.9962) and Greek (Πάρις, 0.9926).
- **Moscow (en):** Retrieves Moscou (French/Portuguese, 0.9945) but misses the primary Cyrillic form Москва (though Московa appears at 0.9941). Also surfaces the data quality issue "Kachua-mokampukur F P School" (0.9874), an OSM institutional name with implausible en/nl language tags.

The results suggest **two distinct phenomena**:

(1) Script family clustering is robust. The model successfully identifies cross-script equivalents in Cyrillic, Arabic, Thai, and Georgian—scripts represented in the training data. The London query's retrieval of Arabic and Georgian variants demonstrates genuine phonetic generalisation beyond the Romance languages explicitly tested.

(2) Romance language under-representation in training data. The consistent failure to retrieve expected Spanish, Italian, Portuguese, and Polish variants across all three queries points to systematic under-sampling rather than model incapacity. The pairwise similarity tests confirm the model *can* match these pairs when directly compared (Rome/Roma: 0.788, Munich/München: 0.878). Their absence from KNN top-100 results suggests they were underweighted during triplet sampling, causing them to occupy regions of the embedding space farther from their Latin-script counterparts than Cyrillic or Arabic equivalents.

This behavior reflects training data composition: while the 27.6M triplet dataset includes toponyms in Spanish, Italian, and Portuguese, the Phase 1 stratified sampling strategy (balanced by lang-script bins with caps at 50k pairs per bin) may have allocated insufficient gradient signal to Romance-language cross-script transfers. The model learns most strongly from high-frequency pairings in the training distribution.

The OSM data quality issue (institutional name retrieval) demonstrates correct model behavior: "Kachua-mokampukur F P School" shares phonetic subsequences with "Moscow" (/mok-/, /mosk-/), so moderate similarity is linguistically appropriate. This highlights the need for data quality filtering at ingestion—institutional names should be tagged with entity type metadata and excluded from geographic feature searches.

Implications for production deployment: KNN retrieval on real gazetteer data must account for high-multiplicity clusters ("London" appears in 69 language variants with identical embeddings, dominating top-k results). Post-processing strategies include: query-time filtering (exclude exact matches to surface cross-script alternatives), script diversity re-ranking, or candidate expansion (retrieve top-100, then filter/rank by additional constraints). The test validates Symphonym as a *candidate retrieval* mechanism rather than a deterministic ranker: embeddings provide phonetically-informed similarity scores, while downstream components apply geographic and entity-type constraints to select among candidates.

Production Implications. The 100% embedding coverage and strong cross-script performance confirm that the hybrid G2P pipeline (Epitran + 102 extensions + Phonikud + CharsiuG2P) successfully addresses all previously identified coverage gaps. The production corpus of 66.9M embedded toponyms enables phonetically-informed search across the full WHG corpus, supporting cross-script reconciliation workflows without requiring language-specific resources or runtime phonetic conversion.

7 Discussion

The evaluation results support Symphonym's central claim: that a character-level neural encoder can learn phonetically meaningful representations across writing systems without requiring explicit phonetic transcription at inference time.

7.1 Key Findings

Version 6 Improvements. The transition from v5 to v6 addresses the critical coverage gaps identified during v5 evaluation. The hybrid G2P strategy—combining 100 new Epitran extension files (assisted by LLMs), Phonikud for Hebrew, and CharsiuG2P for CJK topolects—expands IPA coverage from 50.8% to 53.4% (30.8M of 57.6M training namespace toponyms). The elimination of AnyAscii romanisation for CJK scripts means the Teacher now learns from genuine phonetic representations rather than lossy approximations: Chinese /bei.dʒiŋ/ rather than romanised "beijing", Hebrew /jəruʃaˈlajim/ from unpointed ירושלים rather than absent embeddings. Previously zero-coverage scripts now show substantial IPA generation: Greek (77.4%), Armenian (93.7%), Gujarati (94.9%), Kannada (48.6%). Phase 1 training on the expanded v6 dataset shows val_loss consistently $\sim$1.23$\times$ higher than v5 at equivalent epochs (e.g., 0.0072 vs 0.0054 at epoch 10), reflecting the harder but more linguistically faithful discrimination task. Despite fewer training samples (20.4M vs 24.8M for v5, due to tighter bin-balancing caps), v6's improved script coverage and elimination of romanisation artifacts successfully address the script-specific performance gaps observed in v5 MEHDIE testing. All three training phases completed on NVIDIA L40S GPUs in February 2026.

Cross-script Generalisation. The model achieves strong accuracy on cross-script equivalents in diagnostic tests (27/29 pairs, 93%) and 87.5% Recall@1 on the MEHDIE Hebrew-Arabic benchmark. The two diagnostic failures (Moscow/Москва at 0.845 and Damascus/دمشق at 0.848) fall just below the 0.85 threshold but would still rank correctly in retrieval. The MEHDIE evaluation is particularly notable because the benchmark sources (medieval Hebrew and Arabic geographical texts) are independent of the modern gazetteers used for training, even though Hebrew and Arabic scripts appear in the training data through GeoNames and Wikidata entries. The strong performance suggests the model has learned general cross-script phonetic mappings rather than memorising specific toponyms. This generalisation emerges from the Teacher-Student architecture: the Teacher learns language-specific phonetic mappings from IPA features, then transfers this knowledge to the Student through distillation.

Retrieval vs. Classification. Symphonym excels at *ranking* candidates by phonetic similarity but is not optimised for *classification* at fixed thresholds. The MEHDIE benchmark comparison illustrates this: when evaluated with ranking metrics (R@1, MRR), Symphonym substantially outperforms baselines; threshold-based metrics (F-5) favour methods tuned for specific cutoff points. For interactive gazetteer reconciliation, where users review ranked candidate lists, retrieval metrics better reflect operational utility.

Same-script Limitations. Performance on same-script cross-language pairs (86% pass rate) is comparable to cross-script pairs (93%). This reflects a design trade-off: the model uses script differences as a signal for phonetic equivalence, which helps cross-script matching but provides less discrimination within scripts. For same-script matching, traditional string similarity methods (Levenshtein, Jaro-Winkler) remain effective and can complement Symphonym in a hybrid pipeline.

Hard Negative Training. Phase 3 training sharpens discrimination between orthographically similar names at a modest cost to same-script cross-language pairs. Between epochs 5 and 6, cross-script accuracy improved from 28/29 to 29/29 (Thessaloniki/$\Theta\varepsilon\sigma\sigma\alpha\lambda o\nu\acute{\iota}\kappa\eta$ rose from 0.848 to 0.929), while same-script pairs like London/Londra declined slightly (0.617 to 0.598). This is expected: hard negatives are sampled from the *same* script, teaching the model to be more discriminating within scripts. Since Symphonym's primary goal is cross-script matching—linking "北京" to "Beijing" rather than distinguishing "London" from "Londres"—this trade-off is acceptable. Same-script variants remain linked at the place level through co-location in gazetteer records. Moreover, same-script matching can be effectively handled by traditional string similarity methods (Levenshtein distance, Jaro-Winkler) that complement Symphonym's cross-script capabilities; a hybrid pipeline might apply Symphonym for cross-script candidate retrieval and fall back to edit-distance methods for same-script refinement.

7.2 Operational Implications

Scalability and Deployment. Symphonym embeddings integrate directly with Elasticsearch's HNSW approximate nearest-neighbour indexing, enabling sub-second retrieval over the full WHG corpus of 67M toponyms. Query latency averages 15–50ms including network overhead, making interactive reconciliation workflows practical. Unlike rule-based phonetic algorithms (Soundex, Metaphone), which require exact code matches, embedding similarity enables fuzzy matching with graceful degradation. The Student model's inference cost is minimal: encoding a toponym requires a single forward pass through a 1.7M-parameter network, achievable in <1ms on CPU. This operational efficiency—the fact that Symphonym *actually runs* at scale with acceptable latency—is a key contribution beyond the academic novelty of the embedding approach.

No Runtime Phonetics. The Student encoder requires only raw character input—no language identification, no G2P conversion, no phonetic databases. This simplifies deployment and eliminates failure modes associated with unknown languages or unsupported scripts.

Complementary Methods. Symphonym is designed as one component of a larger reconciliation pipeline. Cross-script candidate retrieval benefits from phonetic embeddings; same-script refinement may use edit distance; final disambiguation incorporates geographic proximity and temporal constraints. The embedding similarity score provides a phonetic prior that downstream components can weigh against other evidence.

Historical Orthographic Variation. Before the standardisation of spelling in many languages (in English, largely during the 18th century), place names were typically written as they sounded to individual scribes, yielding considerable orthographic variation in historical sources. The same city might appear as "Leycester", "Leycestre", "Lecestre", and "Leicester" across different medieval documents. Because Symphonym embeds based on phonetic similarity rather than exact orthography, it naturally groups such variants together: all sound-alike spellings of a place will cluster near the modern canonical form. This enables a practical application beyond cross-script matching: given a historical spelling variant, Symphonym can suggest the most likely modern canonical form by retrieving the nearest neighbours in the embedding space. This is particularly valuable for transcribed historical documents, parish registers, and archival catalogues where inconsistent historical spellings must be linked to standardised gazetteer entries.

7.3 Limitations

Several limitations warrant acknowledgment:

Training Data Bias. Despite the stratified sampling strategy described in Section 4.1, training sources retain biases. GeoNames over-represents populated places with official names, potentially under-representing historical variants. Wikidata's coverage skews toward places of encyclopaedic interest. TGN emphasises art-historically significant locations. Performance on under-represented scripts (e.g., Ethiopic, Myanmar, Tibetan) and mundane places lacking multi-lingual attestations may be weaker.

Tonal Languages. The current architecture does not explicitly model tone, which is phonemically contrastive in Chinese, Vietnamese, Thai, and other languages. The PanPhon features (Section 3.5) encode segmental articulatory properties but not suprasegmental features. In principle, tonal minimal pairs in place names would not be distinguished. However, tonal minimal pairs are rare in practice: Chinese place names like Beijing (北京, běijīng) and Shanghai (上海, shànghǎi) have distinct segmental content that Symphonym captures effectively (0.97–0.99 similarity with romanised forms). The limitation would matter most for hypothetical pairs differing only in tone, which are uncommon in geographic naming.

IPA Coverage. The Teacher's phonetic grounding depends on Epitran's G2P coverage. Of 57.6M training toponyms, 29.0M (50.4%) yielded valid IPA transcriptions and PanPhon embeddings; the remainder contribute only through Phase 2 and 3 character-level training. Coverage varies significantly by script (see Table 2): South Asian scripts (Tamil 90.9%, Telugu 92.6%, Bengali 72.9%) achieve high coverage through Epitran's Indic language support, while scripts like Greek, Hebrew, and Armenian lack G2P models entirely. The strong performance on Georgian (Tbilisi/თბილისი = 0.972) despite Georgian representing only 0.2% of training data suggests the Student learns effective representations through character-level patterns and cross-script transfer.

Confusable Pairs. The model correctly assigns high similarity to phonetically similar strings regardless of semantic relationship. Pairs like Austria/Australia (0.883) and China/Ghana (0.932) receive high scores because they *are* phonetically similar. Disambiguation requires geographic or contextual evidence beyond phonetic similarity.

8 Application to Medieval Personal Name Matching

While Symphonym was designed primarily for toponym matching, its phonetic embedding approach proves equally effective for personal names—particularly in historical contexts where orthographic standardisation is absent. This section presents a case study from the Medieval London Customs Accounts project (University of London), demonstrating Symphonym's utility for general name matching tasks beyond geographic entities.

8.1 The Challenge of Medieval Name Variants

The Medieval London Customs Accounts record thousands of merchants, ship masters, and customs officials, with names transcribed by medieval scribes according to phonetic intuition rather than orthographic rules. The same individual might appear across multiple documents as “Cristofre”, “Cristofer”, “Christofer”, or “Christopher”; a surname might be recorded as “Arnaldson”, “Arnaldsone”, “Arnaldesone”, or “Arnaldisson”. These variants reflect genuine medieval spelling practices rather than transcription errors: pre-standardisation English, Latin, and continental European names were typically written as they sounded to the scribe.

The research objective is to identify these variant forms and group them under canonical representations to enable prosopographic analysis—tracking individuals across multiple customs records, analyzing merchant networks, and studying cross-border trade relationships. This requires clustering phonetically similar name variants without access to modern phonetic algorithms designed for standardised spellings.

8.2 Supervised Learning Framework

Rather than applying Symphonym’s pre-trained toponym model directly, the project uses Symphonym embeddings as *features* within a supervised active learning framework:

1. **Embedding Generation**: 45,139 unique name forms (forenames and surnames) extracted from transcribed customs accounts are encoded using Symphonym’s Student network, producing 128-dimensional phonetic embeddings.
2. **Feature Engineering**: For each candidate name pair, the system computes 11 features:
   - Symphonym cosine similarity (the phonetic embedding distance)
   - Epitran+PanPhon cosine similarity (using 8-bin positional pooling)
   - Jaro-Winkler similarity on normalised forms
   - IPA-level Jaro-Winkler (from Epitran transcriptions)
   - Levenshtein normalised distance
   - Character n-gram cosine similarity
   - Character n-gram Jaccard coefficient
   - Length difference and length ratio
   - IPA n-gram Jaccard coefficient
   - Phonetic edit distance (Levenshtein on IPA)

3. **Active Learning**: A logistic regression classifier is trained on hand-labeled name pairs. The system suggests uncertain pairs (prediction probability near 0.5) for human review, iteratively refining the model.
4. **Cluster Formation**: Once trained, the classifier scores all candidate pairs. Pairs exceeding a threshold are connected via union-find to form clusters of variant spellings.

8.3 Training Results

After 8 rounds of active learning with 323 labeled pairs, the classifier achieved:

- **F1-score**: 0.752
- **Precision**: 0.718 (72% of predicted matches are correct)
- **Recall**: 0.791 (finding 79% of true variant pairs)
- **Accuracy**: 0.762

Logistic regression feature importance analysis revealed that **Symphonym cosine similarity ranked 4th among 11 features** with a coefficient of +1.798, substantially higher than the baseline Epitran+PanPhon cosine similarity (+0.477). The top-ranked features were Levenshtein distance (+5.096), Jaro-Winkler (+4.106), and phonetic edit distance (+4.062), followed by Symphonym.

This demonstrates that Symphonym's learned phonetic representations—trained on modern toponyms from multiple scripts—transfer effectively to *single-script medieval personal names*, despite the domain shift. The model's contribution is particularly notable given that all names in this corpus are Latin-script English: Symphonym's value lies not in cross-script matching (its primary design goal) but in capturing *phonetic similarity patterns* that complement traditional string metrics.

8.4 Example Clusters

The trained system successfully groups historically attested variant spellings:

- **Arnaldson cluster**: Arnaldeson, Arnaldisson, Arnaldson, Arnaldsone (4 members)
- **Andrew cluster**: Andrew, Andrewe (2 members)
- **Ascheburne cluster**: Aschebourne, Ascheburne, Asschebourne, Asscheburne (4 members)
- **Raddewell cluster**: Raddewell, Raddewelle, Radwell, Radwelle, Radewell (5 members)

These clusters align with expert judgment: each group represents phonetically equivalent forms that a medieval scribe might have used interchangeably.

8.5 Implications for Broader Applications

This case study highlights several findings relevant to Symphonym's generalisability:

Cross-Domain Transfer. Symphonym's toponym-trained embeddings transfer to personal names without retraining, suggesting the learned phonetic representations capture general orthography-phonology relationships rather than toponym-specific patterns.

Single-Script Utility. While designed for cross-script matching, Symphonym provides value even within a single script by encoding phonetic similarity patterns that traditional edit-distance metrics may miss. The +1.798 coefficient ($4\times$ stronger than baseline phonetic embeddings) indicates that Symphonym's deep learning approach captures non-linear phonetic relationships.

Hybrid Pipelines. Optimal performance emerged from combining Symphonym with traditional string metrics rather than relying on any single method. The supervised framework allows the model to learn optimal feature weightings for the specific domain.

Historical Orthography. The system's ability to cluster pre-standardisation spelling variants—the original motivation for phonetic embeddings in Section 7—is empirically validated. Medieval names exhibit exactly the kind of phonetically consistent but orthographically inconsistent variation that Symphonym was designed to handle.

8.6 Practical Impact

The resulting name cluster dataset enables researchers to:

- Link merchant records across multiple customs records despite spelling variation
- Identify repeat traders and analyse commercial networks
- Distinguish between genuinely distinct individuals and variant spellings of the same person

- Normalise historical names for database queries while preserving original attestations

This application demonstrates that Symphonym's utility extends beyond the gazetteer reconciliation context for which it was designed. Any scholarly domain involving historical documents with inconsistent orthography—prosopography, onomastics, archival indexing—may benefit from phonetically grounded embeddings.

9 Future Work

While Symphonym achieves strong cross-script performance, several directions remain for future investigation.

9.1 Remaining Technical Challenges

Tonal Language Support. The current architecture does not model tone, which is phonemically contrastive in Chinese, Vietnamese, Thai, and other languages. While tonal minimal pairs are rare among toponyms in practice (see Section 7.3), explicit tone modelling could improve performance on languages where tone carries semantic weight.

Suprasegmental Features. PanPhon encodes articulatory properties of individual segments but not prosodic features like stress patterns, syllable structure, or intonation. Toponyms that differ primarily in stress placement (e.g., certain Greek names with movable accent) may not be adequately distinguished.

Historical Orthography. The model is trained on modern standardised spellings. Performance on pre-standardisation documents—where spellings reflect local pronunciation and writer idiosyncrasy—remains to be systematically evaluated. The phonetic grounding should help (sound-alike variants cluster together), but the character vocabulary may lack rare historical graphemes.

Expanded Script Coverage. The current system covers 20 script categories, but scripts like Ethiopic, Myanmar, and Tibetan are absent or underrepresented. While v6 has addressed critical gaps for Hebrew and Chinese topolects via neural G2P integration, expanding coverage to further scripts requires both training data (multi-script gazetteer entries) and additional G2P resources.

9.2 Architectural Extensions

Multi-Task Learning. The current pipeline treats toponym matching as a standalone task. Joint training with related tasks—language identification, script detection, transliteration—could provide additional supervisory signal and improve representations.

Contextual Embeddings. Symphonym embeds individual toponyms without context. For ambiguous names ("Paris" in France vs. Texas), incorporating surrounding text or geographic constraints could improve disambiguation. This would require architectural changes to accept longer input sequences.

Incremental Model Updates. As the WHG index grows and new authorities are ingested, the model should be updatable without full retraining. Techniques like experience replay or elastic weight consolidation could enable continual learning while preserving performance on established script pairs.

9.3 Deployment and Integration

Hybrid Retrieval Pipeline. The current evaluation treats Symphonym in isolation. Production deployment would benefit from a hybrid approach: phonetic embeddings for cross-script candidate retrieval, traditional string metrics for same-script refinement, and geographic filtering for final disambiguation. Systematic evaluation of such hybrid pipelines remains future work.

User Studies. The ultimate measure of success is whether Symphonym helps researchers link historical sources. User studies with digital humanities practitioners would provide qualitative feedback on false positive/negative rates in realistic workflows.

Benchmark Expansion. The MEHDIE benchmark provides valuable Hebrew-Arabic evaluation data. Additional benchmarks spanning other script pairs (Latin-Chinese, Cyrillic-Georgian, Devanagari-Arabic) would enable more comprehensive evaluation of cross-script generalisation.

10 Conclusion

I have presented Symphonym, a neural embedding system for cross-script toponym matching that places phonetically similar names near each other in a unified 128-dimensional space regardless of writing system. The three-phase training curriculum—phonetic feature learning, student-teacher distillation, and hard negative fine-tuning—enables the model to generalise across scripts without requiring phonetic resources at inference time. Training completed February 2026 on NVIDIA L40S GPUs, with all three phases successfully converging. The v6 implementation processes CJK, Hebrew, Greek, Armenian, and other scripts in their native writing systems rather than through romanisation, using appropriate G2P methods (CharsiuG2P for CJK topolects and Korean, Phonikud for Hebrew, extended Epitran for 100+ language-script pairs) to generate IPA transcriptions for Teacher training.

Evaluation demonstrates strong performance:

- **93%** accuracy on cross-script diagnostic pairs (63/68), spanning Latin, Cyrillic, Greek, Arabic, Chinese, Korean, Hebrew, Devanagari, and Georgian scripts
- **87.5%** Recall@1 and **MRR of 0.923** on the MEHDIE Hebrew-Arabic benchmark—comprising medieval toponyms from sources independent of training data—outperforming Levenshtein (81.5% R@1) and Jaro-Winkler (78.5% R@1) baselines

The system addresses a practical need: linking place references across languages, scripts, and historical periods without requiring language-specific rules or phonetic lookup tables. By learning directly from the statistical structure of multi-script toponyms attested in linked authority datasets, Symphonym captures phonetic correspondences that would be difficult to enumerate manually.

Beyond toponyms, the personal name matching case study (Section 8) demonstrates successful transfer to a substantially different domain: single-script medieval personal names with historical orthographic variation. In a supervised active learning framework combining 11 phonetic and string-similarity features, Symphonym cosine similarity ranked 4th in importance with a coefficient 4× stronger than baseline phonetic embeddings, contributing to F1=0.752 on medieval name variant clustering. This validates the hypothesis that Symphonym's learned representations capture general orthography-phonology relationships applicable to any domain with phonetically consistent but orthographically variable names—including prosopography, archival indexing, and historical onomastics.

Symphonym is to be deployed in the World Historical Gazetteer, enabling fuzzy phonetic search and reconciliation across 67M+ toponyms from antiquity to the present. The trained models and training code are released to support further research in historical toponym linking, cross-lingual named entity matching, and phonetic name clustering across diverse scholarly applications.

Data Availability

The training data are derived from publicly available sources: GeoNames (https://www.geonames.org/), Wikidata (https://www.wikidata.org/), and the Getty Thesaurus of Geographic Names (https://www.getty.edu/research/tools/vocabularies/tgn/). The MEHDIE benchmark used for evaluation is available via the supplementary file link at https://link.springer.com/article/10.1007/s1
025-09812-9. Extracted training datasets and pre-computed embeddings will be made available upon publication.

Code Availability

Training code and inference utilities are available at: https://github.com/WorldHistoricalGazetteer/indexing. The repository includes technical documentation of the HDBSCAN clustering methodology used for positive pair generation. The system integrates with Elasticsearch for scalable deployment.

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